Rakan Eyad

Software Engineer

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SUMMARY

Mechatronic engineering graduate with full-stack software engineering experience. Built a production micro-SaaS with telephony, AI, and billing, a full-stack payments platform, and an open-source Python engine profiled to a 58x speed-up and shipped as an installable SDK.

EXPERIENCE

System Engineer — ALBON Nov 2024 — Jan 2026

- Developed and integrated hardware, software, and mechanical components for a prototype control system: migrated the architecture from a microcontroller to a Raspberry Pi for real-time data processing, integrated sensor arrays, and engineered the system's electrical junction box to cut power consumption and keep the hardware reliable.

Engineering Intern — Escape The Room Sep 2021 — Nov 2021

- Designed mechanical puzzles and implemented engineering concepts in real-world, client-facing products.

Testing Technician Intern — CloudAk May 2021 — Aug 2021

- Supported server system testing and maintenance in data-centre environments.

FEATURED PROJECTS

Tradie Text — Micro-SaaS · in production

- Built a production micro-SaaS that recovers missed calls for trades businesses using Twilio telephony and a GPT-4o-mini SMS conversation engine.
- Built multi-tenant infrastructure end-to-end: Next.js 15 + TypeScript, Supabase Postgres with row-level security, Redis-backed idempotency/rate limiting, and Stripe usage-based billing.
- Hardened the system solo with 276 CI tests, Playwright e2e coverage, a threat model, and incident runbooks.

FUN.D — Full-stack web app

- Built a full-stack crowdfunding platform: a Flask REST API (JWT auth, role-based admin) and a React 19 + Vite frontend.
- Implemented an idempotent Stripe Elements payment flow end-to-end and resolved N+1 queries, upload validation, and CI across both stacks.

Conjure — Open source · Python

- Built a real-time computer-vision engine classifying 9 hand/face gestures via MediaPipe landmarks, rendering effects at 60 FPS with unit tests and CI.
- Profiled and optimized the render pipeline from 162 ms to 2.8 ms per frame (58x speed-up) through allocation discipline.

Conjure SDK — Open source · Python

- Extracted the engine into a pip-installable SDK with an event-driven API for driving games, overlays, or hardware from gesture events.
- Designed two install tiers (numpy-only and full webcam) with deterministic, webcam-free tests.

SKILLS

Software engineering: Python / TS, React · Next.js · Flask, Postgres · Supabase · Redis, Stripe · Twilio integration, pytest · Vitest · Playwright · CI, git · GitHub · profiling

Working style: Test-driven & methodical, Documents thoroughly, Led teams to delivery, Client-facing comms, Self-taught: ROS2, FPGA, Isaac, Adaptable across the stack

EDUCATION

B.E. (Honours), Mechatronic Engineering

University of Sydney · 2022 — 2026

EWAM (Engineering Weighted Average Mark) 76% — Distinction